

Project: TourismGov - Public Tourism & Heritage Management System

1. Introduction

TourismGov is a web-based platform designed for tourism ministries, heritage boards, and government agencies to manage tourism programs, heritage sites, and compliance. It enables citizens and tourists to access information, book services, and participate in cultural events, while administrators and officers can monitor compliance, manage tourism projects, and generate reports.

The system supports workflows for tourist registration, heritage site management, event scheduling, resource allocation, compliance monitoring, and analytics. It ensures transparency and accountability by maintaining audit trails, dashboards, and performance metrics across tourism governance.

Actors / Users

- **Citizen/Tourist:** Registers, books tours, tracks heritage site visits.
- **Tourism Officer:** Validates bookings, manages sites/events, updates status.
- **Program Manager:** Oversees tourism programs, monitors budgets, tracks performance.
- **Administrator:** Configures workflows, manages users, oversees reporting.
- **Compliance Officer:** Ensures adherence to tourism policies, audits records.
- **Government Auditor:** Reviews compliance reports, monitors tourism program utilization.

2. Module Overview

- 2.1 Identity & Access Management
- 2.2 Tourist Registration & Profile Management
- 2.3 Heritage Site & Monument Management
- 2.4 Event & Tour Scheduling
- 2.5 Tourism Program & Resource Management
- 2.6 Compliance & Audit Management
- 2.7 Reporting & Analytics
- 2.8 Notifications & Alerts

3. Architecture Overview

- **Frontend:** Angular or React for responsive dashboards.

- **Backend:** REST API-based microservices (Identity, Tourists, Sites, Events, Programs, Compliance, Reports).
- **Database:** Relational DB (MySQL/PostgreSQL/SQL Server).
- **Deployment:** Cloud/on-prem with API gateway, WAF, centralized logging.

4. Module Wise Design

4.1 Identity & Access Management

Features

- Secure authentication and role-based access control.
- Audit logging of all actions.

Entities

- **User**(UserID, Name, Role [Tourist/Officer/Manager/Admin/Compliance/Auditor], Email, Phone, Status)
- **AuditLog**(AuditID, UserID, Action, Resource, Timestamp)

4.2 Tourist Registration & Profile Management

Features

- Registers tourists and maintains profiles.
- Validates identity and travel documents.

Entities

- **Tourist**(TouristID, Name, DOB, Gender, Address, ContactInfo, Status)
- **TouristDocument**(DocumentID, TouristID, DocType [IDProof/Passport], FileURI, UploadedDate, VerificationStatus)

4.3 Heritage Site & Monument Management

Features

- Manage heritage sites and monuments.
- Track preservation activities.
- Monitor site status.

Entities

- **HeritageSite**(SiteID, Name, Location, Description, Status)
- **PreservationActivity**(ActivityID, SiteID, OfficerID, Description, Date, Status)

4.4 Event & Tour Scheduling

Features

- Schedule cultural events and tours.
- Track bookings and participation.

Entities

- **Event**(EventID, SiteID, Title, Location, Date, Status)
- **Booking**(BookingID, TouristID, EventID, Date, Status)

4.5 Tourism Program & Resource Management

Features

- Manage tourism programs (festivals, campaigns).
- Allocate resources (funds, venues, staff).
- Monitor program outcomes.

Entities

- **TourismProgram**(ProgramID, Title, Description, StartDate, EndDate, Budget, Status)
- **Resource**(ResourceID, ProgramID, Type [Funds/Venue/Staff], Quantity, Status)

4.6 Compliance & Audit Management

Features

- Ensure adherence to tourism policies.
- Maintain compliance records.
- Conduct audits with findings.

Entities

- **ComplianceRecord**(ComplianceID, EntityID, Type [Site/Event/Program], Result, Date, Notes)
- **Audit**(AuditID, OfficerID, Scope, Findings, Date, Status)

4.7 Reporting & Analytics

Features

- Generate dashboards for sites, events, programs, and compliance.
- Provide analytics on tourism program efficiency.

Entities

- **Report**(ReportID, Scope [Site/Event/Program], Metrics, GeneratedDate)

4.8 Notifications & Alerts

Features

- Send alerts for bookings, event updates, compliance deadlines.
- Notify tourists about program updates.

Entities

- **Notification**(NotificationID, UserID, EntityID, Message, Category [Site/Event/Program/Compliance], Status, CreatedDate)

5. Deployment Strategy

- **Local**: Angular/React frontend, Spring Boot/.NET Core backend, local DB.
- **Production**: Cloud/on-prem deployment, API gateway, WAF, centralized logging.

6. Database Design

Tables & Attributes

- **User**(UserID, Name, Role, Email, Phone, Status)
- **AuditLog**(AuditID, UserID, Action, Resource, Timestamp)
- **Tourist**(TouristID, Name, DOB, Gender, Address, ContactInfo, Status)
- **TouristDocument**(DocumentID, TouristID, DocType, FileURI, UploadedDate, VerificationStatus)
- **HeritageSite**(SiteID, Name, Location, Description, Status)
- **PreservationActivity**(ActivityID, SiteID, OfficerID, Description, Date, Status)
- **Event**(EventID, SiteID, Title, Location, Date, Status)
- **Booking**(BookingID, TouristID, EventID, Date, Status)
- **TourismProgram**(ProgramID, Title, Description, StartDate, EndDate, Budget, Status)
- **Resource**(ResourceID, ProgramID, Type, Quantity, Status)
- **ComplianceRecord**(ComplianceID, EntityID, Type, Result, Date, Notes)
- **Audit**(AuditID, OfficerID, Scope, Findings, Date, Status)
- **Report**(ReportID, Scope, Metrics, GeneratedDate)
- **Notification**(NotificationID, UserID, EntityID, Message, Category, Status, CreatedDate)

7. User Interface Design

- **Tourist Portal**: Event booking, heritage site information, notifications.
- **Tourism Officer Dashboard**: Booking validation, site management, compliance tracking.

- **Program Manager Console:** Program oversight, resource monitoring, reporting.
- **Administrator Panel:** Workflow setup, compliance monitoring, reporting.
- **Compliance Officer Console:** Policy monitoring, audits, compliance reporting.
- **Government Auditor Dashboard:** Tourism program review, compliance monitoring.

8. Non-Functional Requirements

- **Performance:** Handle 150,000 concurrent users across tourism networks.
- **Security:** Role-based access, encrypted data storage, immutable audit logs.
- **Scalability:** Support nationwide rollout across multiple tourism programs.
- **Availability:** 99.9% uptime.
- **Maintainability:** Modular microservices, API versioning, automated migrations.
- **Observability:** Centralized logging, KPIs (booking success rate, event participation rate, compliance adherence).

9. Assumptions & Constraints

- Initial rollout for a single tourism program before nationwide expansion.
- Notifications limited to in-app and SMS/email alerts.
- Fully implementable using Java Spring Boot/ASP.NET Core, Angular/React, and relational DB.